Log Book Entry 1.2

NB2420 Electronic Instrumentation · Manual 1.2 · Oscilloscope and filters

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

Coupling Tasks B1, A1 and I1, leading to C1 *Square wave, 4 U_{pp}, 25 Hz, with a −2 V DC offset. Predict each average as homework, then measure it at the bench.*

Duty cycle	U_{avg} predicted A1	U_{avg} measured I1	DC coupling shows	AC coupling shows
50 %				
75 %				
25 %				

Triggering Tasks B2, A2 and I2, leading to C2

Trigger setting	Predicted display	Observed display
Level inside the signal range		
Level above the signal peak		
Slope changed rising to falling		

Filter behaviour Task B3,A3,I3, leading to C3: 100 kΩ with 47 nF

	Predicted	Simulated	Measured
Behaviour (low or high pass)			
−3 dB point			

Frequencies you swept, and the amplitude ratio at each:

Graph *Plot the measured amplitude ratio against frequency and mark the −3 dB point.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: